Fiscal Year:	FY 2005	**Task Last Updated:**	FY 02/28/2006
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Project Title:	3D Growth Effects on Drug Resistance in Human Ovarian Tumor Cells		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
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Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77030	**Congressional District:**	18
Comments:			
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One of the problems in studying drug resistance in human cancer is the lack of suitable systems which reproducibly model conditions occurring in vivo. Clinical drug resistance in vivo generally occurs concomitantly with, or shortly thereafter, the administration of chemotherapy and very often manifests as resistance to multiple agents even in only single agent therapy is given. The nature of this rapid onset of drug resistance is poorly understood. The purpose of this research is to culture human ovarian tumor cells three- dimensionally in the NASA-developed High Aspect Ratio Rotating-Wall Vessel (HARV), which provides a model that is more biologically representative of conditions occurring in vivo. This model will be utilized study chemotherapeutic drug resistance developing as a function of three-dimensional growth The task objective of this research is to examine specific cell cycle changes and apoptosis associated with the modulation of human ovarian tumor cell drug resistance during three dimensional cell culture in the

Task Description:	HARV. In contrast to acquired resistance, mechanisms underlying the rapid onset of drug resistance in human cancer are not well elucidated. It has been suggested that this type of resistance is not unicellular in nature but is, in fact, multicellular and a function of cell-cell and cell-matrix interactions and three-dimensional architecture. Thus, the development of tumor models using three-dimensional cell culture technology affords great potential not only for improving cancer therapy, but also for providing a superior model for gaining a better understanding of methods to overcome the development of rapid resistance. We have taken this one step further in the present project year, by analysis of gene expression patterns in drug treated three dimensional constructs. This model therefore facilitates a high fidelity testing platform for response to therapeutics, and the use of gene expression profiling to ascertain patterns to be used as biomarkers of drug resistance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The earth based benefits from this research derived from the ability to provide a model more human ovarian tumor growth that is more biologically representative than other currently available models. In vivo, ovarian tumor cells spread from the primary lesion via exfoliation. These peeled off masses of cells float as non-vascularized small aggregates in ascitic fluid, where they subsequently attach to peritoneal surfaces and in doing so disseminate throughout the peritoneal cavity. Growth as three dimensional constructs in the HARV is a particularly relevant model for the study of metastatic human ovarian cancer, in that maintenance of multicellular aggregate structures in a suspended culture environment simulates closely the aspects of both solid and suspension tumor growth that occurs in this disease in vivo.The cells grown in three dimensional constructs exhibit significantly elevated drug resistance, as a function of their three dimensional multicellular architecture, and also exhibit significantly reduced proliferative phase fractions, again mirroring what occurs in large, bulky ovarian tumors in vivo. Studies are underway to evaluate the ability of this model to serve as a testing platform to ascertain the specific gene expression "fingerprint" patterns predictive of clinical response rate.
Task Progress:	International Space Station (ISS) data remained under active investigation at Johnson Space Center (JSC). Human ovarian mixed mullerian tumor cells (LN1 cells) were delivered to ISS on STS-105 and grown on Cytodex microcarriers for 14 days in microgravity during Expedition 3. Following termination of the study, samples were preserved in 'RNALater' and were stored on-orbit for a period of 3 months. Following sample return on STS-108, the cells were examined for antigenic stability after removal of RNA. Using Western blot analysis, determination of the levels of vimentin and epithelial membrane antigen were evaluated and compared to ground controls. Using bioluminescence (Luminex Bioanalyzer), the levels of the secreted cytokines interleukin (IL) -6 and IL-8 were evaluated in flight and ground samples. Over the course of the 14 day study, ground control and flight LN1 cell samples showed a similar staining pattern over time with antibodies to vimentin and epithelial membrane antigen. With time in culture, the cells produced increasing amounts of IL-6 and IL-8 both on ground and in flight. The flight samples exhibited less cytokine production overall, however, and also less cell growth, as determined by monitoring of oxygen consumption and glucose utilization (this was accomplished via iSTAT-portable clinical blood gas analyzer) . It was determined that this result could be due to microgravity and/or additional operational parameters of the flight environment in which the cells were grown on-orbit. Relocation of my laboratory to Baylor College of Medicine facilitated a close working relationship with the intramural scientists at JSC which facilitated greatly the ability to present these data and publish the findings during this year. Analyses on the remaining samples from this work, which have been stored at JSC, will continue and will hopefully facilitate uncovering of more information which will aid in increasing our understanding of the changes in cellular growth and physiology of human ovarian tumor cells grown as multicellular constructs. Studies are ongoing with 3D constructs of human ovarian carcinoma (OVCAR-3) cells to serve as a model for development of gene expression profiles for use as biomarkers of drug resistance applicable to establishment of predictive models for determining clinical response rate of human tumors. Initial studies characterized the expression of 7,820 genes per sample (N=10) in control and drug therapy groups, comprised of paclitaxel and carboplatin, two drugs routinely utilized to treat human disease. Data were evaluated using a Bayesian binary regression model with Metagene analysis (wherein groups of genes are identified according to factors associated with phenotype). This analysis revealed that alpha tubulin (genes: TUBA1, TUBA2, TUBA3) and beta tubulin (genes: TUB3, TUBB4, TUBB5, beta2) were upregulated in paclitaxel treated cells, together with intracellular hyaluronic acid binding protein (HMMR) and cyclin A (CCNA2). More extensive analyses have now been initiated using a gene expression predictive model, wherein results in expression profiles obtained in vitro in the 3D treated tumor constructs are being compared with results obtained from patient tumors exposed to paclitaxel or carboplatin in vivo to determine predictive accuracy of the model system for assessing fingerprints of drug resistance gene profiles observed in vivo. These analyses are being performed with collaborators at the Duke University Microarray Facility (who have pioneered the Metagene analysis technique). This work is being prepared for presentation, after which the data will be compiled and published. Of note, this year a new industry contact was made with ImClone, Inc. (New York, NY), makers of Erbitux, for testing the 3D model for the effects of this next generation anti-growth factor receptor agent on her-2/neu. The plans are to utilize data from these studies to help guide the design of patient trials of this agent for the treatment of human ovarian cancer.
Bibliography Type:	Description: (Last Updated: 08/09/2012)
Abstracts for Journals and Proceedings	Becker JL, Fiedler ER. "The human in human space exploration. Benefits of the human presence in space." 15th IAA Humans in Space Symposium, Graz, Austria, May 22-26, 2005. 15th IAA Humans in Space Symposium, Graz, Austria, May 22-26, 2005. , May-2005
Articles in Peer-reviewed Journals	Hammond DK, Becker JL, Elliott TF, Holubec K, Baker TL, Love JE. "Antigenic protein in microgravity-grown human mixed Mullerian tumor cells (LN1) preserved in RNA stabilizing agent." Gravit Space Biol. 2005 Jun;18(2):99-100. http://gravitationalandspacebiology.org/index.php/journal/article/view/357/360 , Jun-2005